Chapter 1

Verification Cone Calorimeter Gpyro

Introuduction ...

1.1 Reference Cone Calorimeter Configuration

This validation case aims to assess the ability of Gpyro to reproduce a standard one-dimensional pyrolysis scenario representative of cone calorimeter tests, and to compare the results with those from the reference codes Thermakin and FDS, as detailed in the study [1].

A fictitious but physically coherent material, referred to as **Material A**, is used. Two independent first-order chemical reactions are modeled using the kinetic parameters detailed in Table 1.1, and the thermophysical properties of all solid species are provided in Table 1.2.

The physical configuration consists of a 6 mm thick layer of Material A placed atop a 12 mm Kaowool insulation layer. A constant external heat flux of 50 kW/m^2 is applied to the top surface. Convective heat transfer is imposed with a heat transfer coefficient of $h_c =$ 8.2 W/m^2·K and an ambient temperature of 25 °C. The bottom surface exchanges heat with the environment solely via thermal radiation. Figure 1.1 illustrates the configuration.

To ensure equivalence with the reference codes, the following modeling assumptions are adopted:

- Gas-phase contributions to energy conservation are neglected.
- Gas release is assumed to be instantaneous (no gas transport).
- Reaction enthalpies are treated as constant.
- No material deformation is modeled (densities chosen to preserve volume).

A uniform spatial discretization of 0.02 mm and a time step of 0.05 s are used. Numerical convergence is verified under these settings.

Table 1.1: Chemical kinetics parameters

Reaction	A_k (s^{-1})	E_k (kJ/mol)	n_k (-)	ΔH_k (J/kg)	θ_k (kg/kg)
A $\rightarrow \theta_1$ B + $(1-\theta_1)$ Gas$_1$	9.5×10^{20}	249	1	1.7×10^5	0.44
B $\rightarrow \theta_2$ C + $(1-\theta_2)$ Gas$_2$	5.5×10^{11}	192	1	1.2×10^6	0.47

Table 1.2: Thermophysical properties of solid species

Material	k (W/m·K)	c_p (J/kg·K)	ρ (kg/m^3)	α (m^{-1})	ε (-)
Material A	$0.17 + 3.10^{-4}T$	1550	1430	∞	1.0
Material B	$0.24 + 8.10^{-4}T$	1720	629.2	∞	1.0
Material C	$0.36 + 18.10^{-4}T$	780	265.7	∞	1.0
Kaowool	0.058	1070	256	∞	0.95

The results obtained using Gpyro are compared to those from Thermakin and FDS. Figure 1.2 presents the comparison of the Mass Loss Rate (MLR) profiles. Figure 1.3 illustrates the evolution of temperature at various depths, while Figure 1.4 shows the time-dependent evolution of species concentrations at a depth of 3 mm.

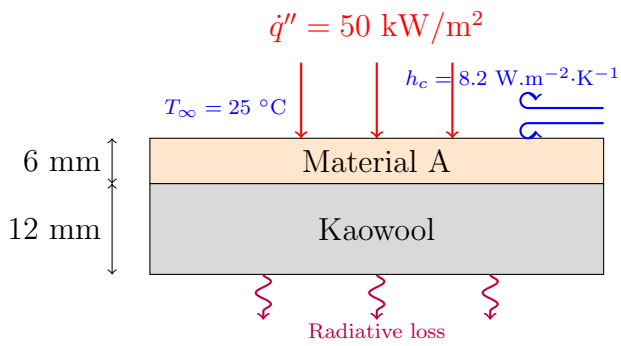

Figure 1.1: Configuration schematic

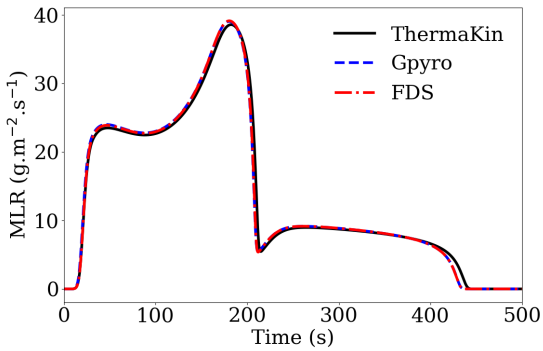

Figure 1.2: Mass Loss Rate (MLR) comparison

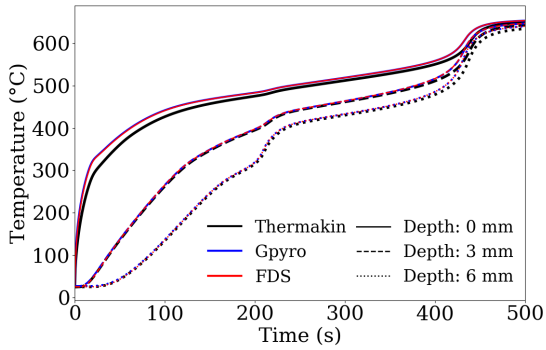

Figure 1.3: Temperature evolution at different depths

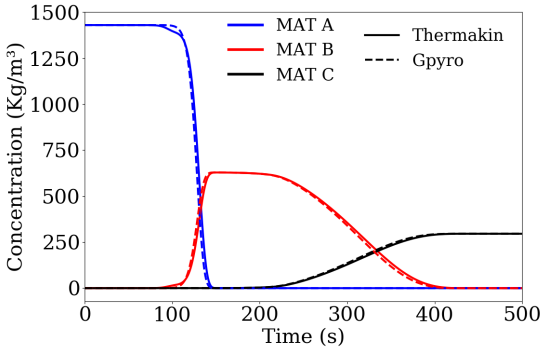

Figure 1.4: Species concentration evolution at 3 mm depth

1.2 Temporal Convergence Verification

This verification case assesses the temporal convergence behavior of the Gpyro solvers in a cone calorimeter configuration. The reference simulation corresponds to the `reference_cc` setup. A one-dimensional uniform spatial discretization of $\Delta z = 0.1\,\text{mm}$ is used for all simulations, while the time step Δt is varied from 10 s to 0.0005 s.

The mass loss rate (MLR) curves obtained for each time step are presented in Figure 1.5, and Table 1.3 summarizes the corresponding mean errors and CPU times.

For time steps smaller than 1 s, the solution exhibits convergence. For larger time steps, deviations appear due to temporal discretization errors. However, because the 1D thermal solver is implicit and thus unconditionally stable, these deviations remain bounded and manifest as oscillations rather than instabilities. Moreover, the species conservation equation is implemented in a conservative form. As a result, even with large time steps, the total mass loss remains consistent across simulations—an important property for practical robustness.

The temporal error is evaluated by using the simulation with the smallest time step (0.0005 s) as a reference. The error is computed as the mean absolute difference between the MLR curves of each simulation and the reference, integrated over time.

Given the first-order nature of the temporal scheme, the error is expected to scale linearly with Δt, i.e., $\mathcal{O}(\Delta t^1)$. This is confirmed by the log-log plot of the error versus time step shown in Figure 1.6, where a linear regression is performed to estimate the convergence order.

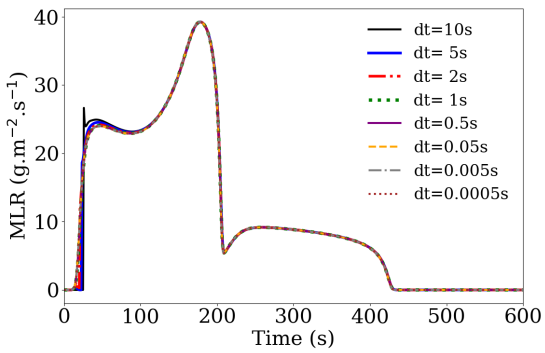

Figure 1.5: Mass loss rate profiles for different time steps.

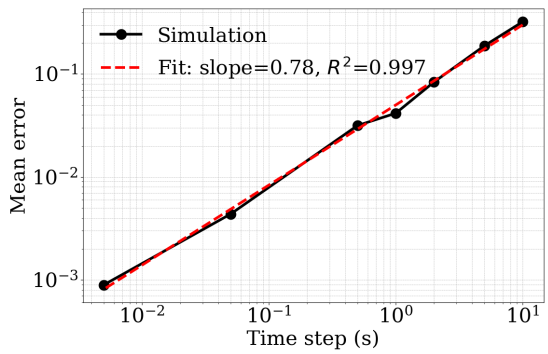

Figure 1.6: Mean error vs. time step.

Table 1.3: Mean error and CPU time for different time step sizes.

Δt (s)	Mean Error	CPU Time (s)

1.3 Spatial Convergence Verification

This verification case assesses the spatial convergence behavior of the `Gpyro` solver in a cone calorimeter configuration. All simulations are run using a fixed time step $\Delta t = 0.05\,\mathrm{s}$, while the spatial grid size Δz is varied from 1 mm to 0.0001 mm.

Figure 1.7 presents the mass loss rate obtained for different spatial discretizations, and Table 1.4 summarizes the corresponding mean errors and CPU times.

A log-log plot of the error versus Δz is shown in Figure 1.8. The linear regression indicates that the numerical scheme exhibits a spatial convergence rate close to $\mathcal{O}(\Delta z^1)$.

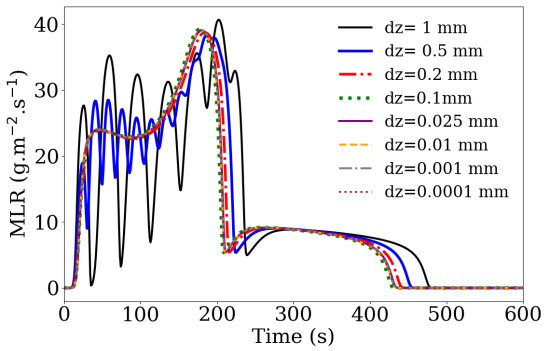

Figure 1.7: Mass loss rate profiles for different grid sizes.

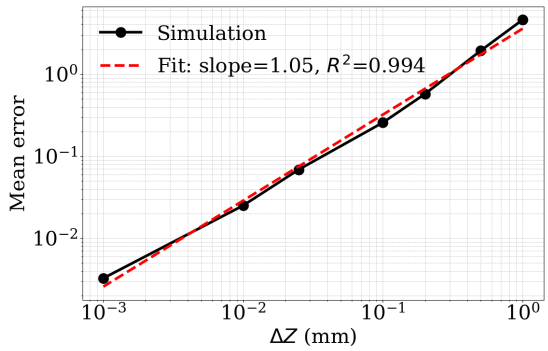

Figure 1.8: Mean error vs. grid size Δz.

Table 1.4: Mean error and CPU time for different spatial grid sizes.

Δz (mm)	Mean Error	CPU Time (s)

1.4 Extension to 2D and 3D Configurations

To further evaluate the robustness and consistency of the solver, the reference one-dimensional cone calorimeter configuration is extended to two- and three-dimensional domains. The domain is expanded to 10 cm $\times$ 10 cm in the x and y directions, while maintaining a vertical thickness of 18 mm identical to the 1D case.

The computational mesh is composed of 20 uniform cells in both the x and y directions, and 180 cells in the z direction. This corresponds to a spatial resolution of $\Delta x = \Delta y =$ 5 mm and $\Delta z = 0.1$ mm, which is 50 times finer along the vertical axis. A constant time step of $\Delta t = 0.5$ s is employed throughout the simulation.

To avoid the stability constraints typically imposed by the fine vertical resolution, the numerical scheme is configured to be implicit along the z-axis and explicit in the x and y directions. This is achieved by activating the option `SOLVER_MAIN_DIRECTION='Z'`.

In order to ensure theoretical equivalence with the one-dimensional configuration, adiabatic boundary conditions are imposed on all lateral surfaces (in the x and y directions). Under these assumptions, the 2D and 3D configurations are expected to reproduce the same thermal and chemical behavior as the 1D case.

The comparison of the Mass Loss Rate (MLR) obtained from the 1D, 2D, and 3D simulations is presented in Figure 1.9. The consistency between the results confirms the proper extension of the solver to multidimensional domains without introducing numerical bias.

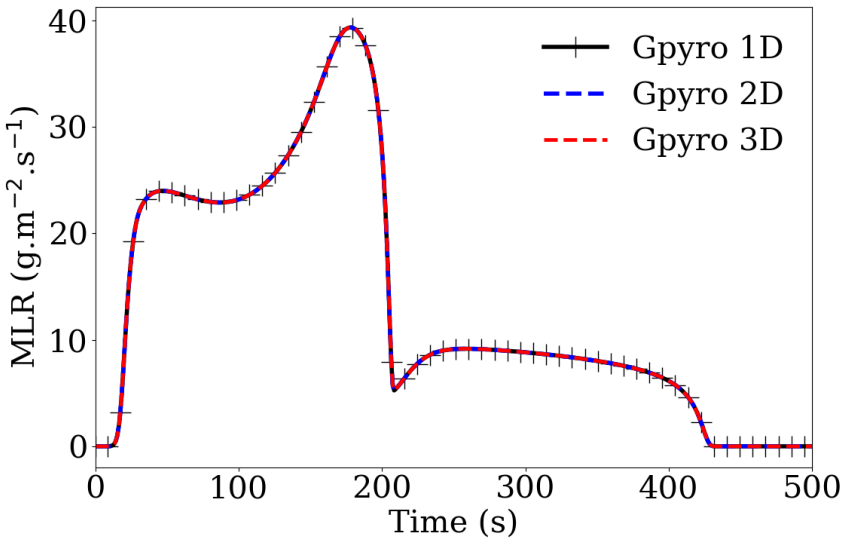

Figure 1.9: Comparison of Mass Loss Rate (MLR) between 1D, 2D, and 3D configurations

1.5 3D Cone Calorimeter with Lateral Convection

This case extends the reference cone calorimeter configuration into three dimensions by introducing natural convection on the lateral x and y surfaces, in addition to the top-applied heat flux. This setup induces multidimensional effects in the degradation front propagation, resulting in a more realistic physical behavior.

The domain consists of a 6 cm thick slab of **Material A** atop a 12 cm Kaowool insulation layer, with lateral dimensions of 10 cm × 10 cm. A vertical mid-plane at $x = 5$ cm (shown in gray in Figure 1.10) is used to visualize the pyrolysis front using Smokeview. Figure 1.11 shows the reaction rate distribution in this plane at various time steps.

The computational grid uses uniform spacing: $\Delta z = 0.1$ mm to resolve vertical gradients, and $\Delta x = \Delta y = 5$ mm laterally. A fixed time step of $\Delta t = 0.5$ s is applied. The solver is configured with `SOLVE_MAIN_DIRECTIONS = Z`.

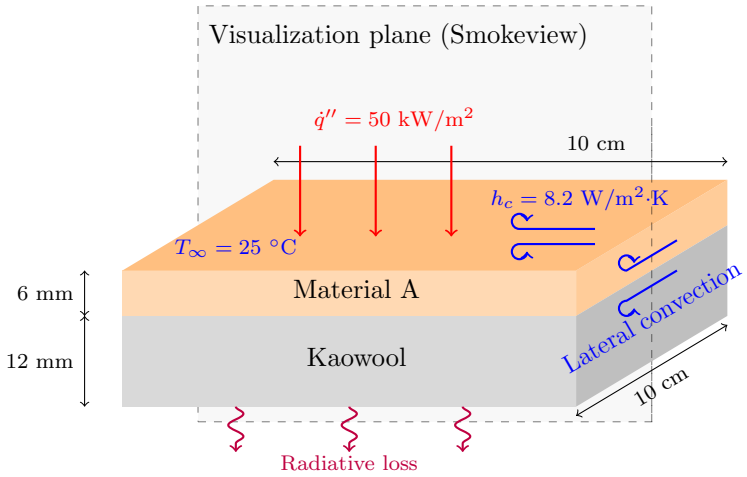

Figure 1.10: Three-dimensional configuration.

smokeview_3D_conv.png

Figure 1.11: Reaction rate distribution in the mid-height plane YZ at different times.

Bibliography

[1] Youssouf Abdelhafiz, Abdenour Amokrane, Gerald Debenest, and Serge Bourbigot. Comprehensive comparison of pyrolysis codes with a focus on their energy conservation modelling. *To be submitted*, 2025.